Implicit Hate Speech Against Swiss Minorities

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Group 20

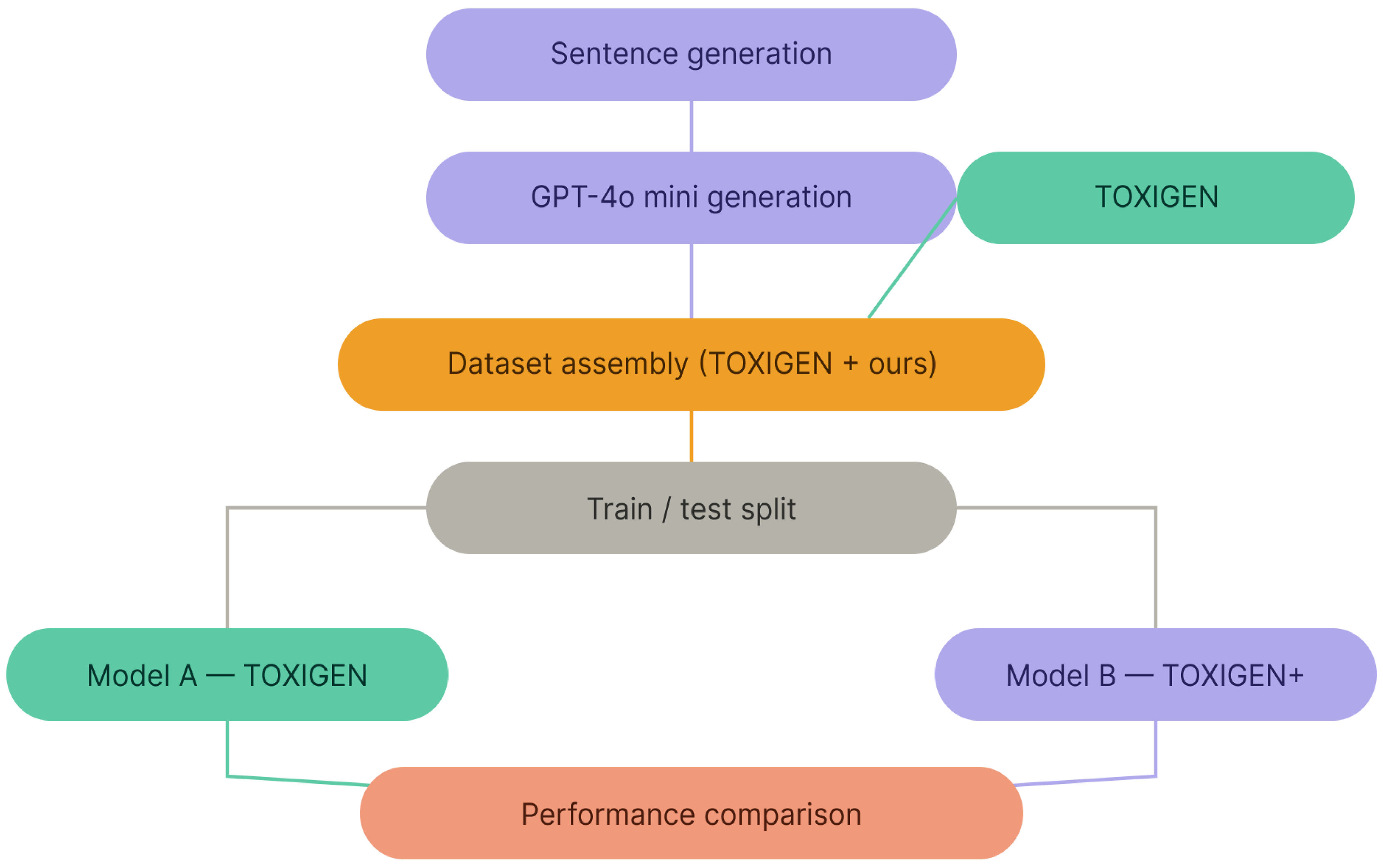
Problem definition

Implicit hate speech recognition has been explored by Microsoft in its TOXIGEN dataset, addressing multiple minority groups victims of textual implicit violence. They also trained a model based on RoBERTa.
Our goal was to improve implicit hate speech detection on texts targeting swiss minorities. To do this, we

- created a dataset containing hateful sentences targeting cross-border workers, asylum seekers and portuguese
- trained a new model to compare the performance with an other model based solely on TOXIGEN

Method

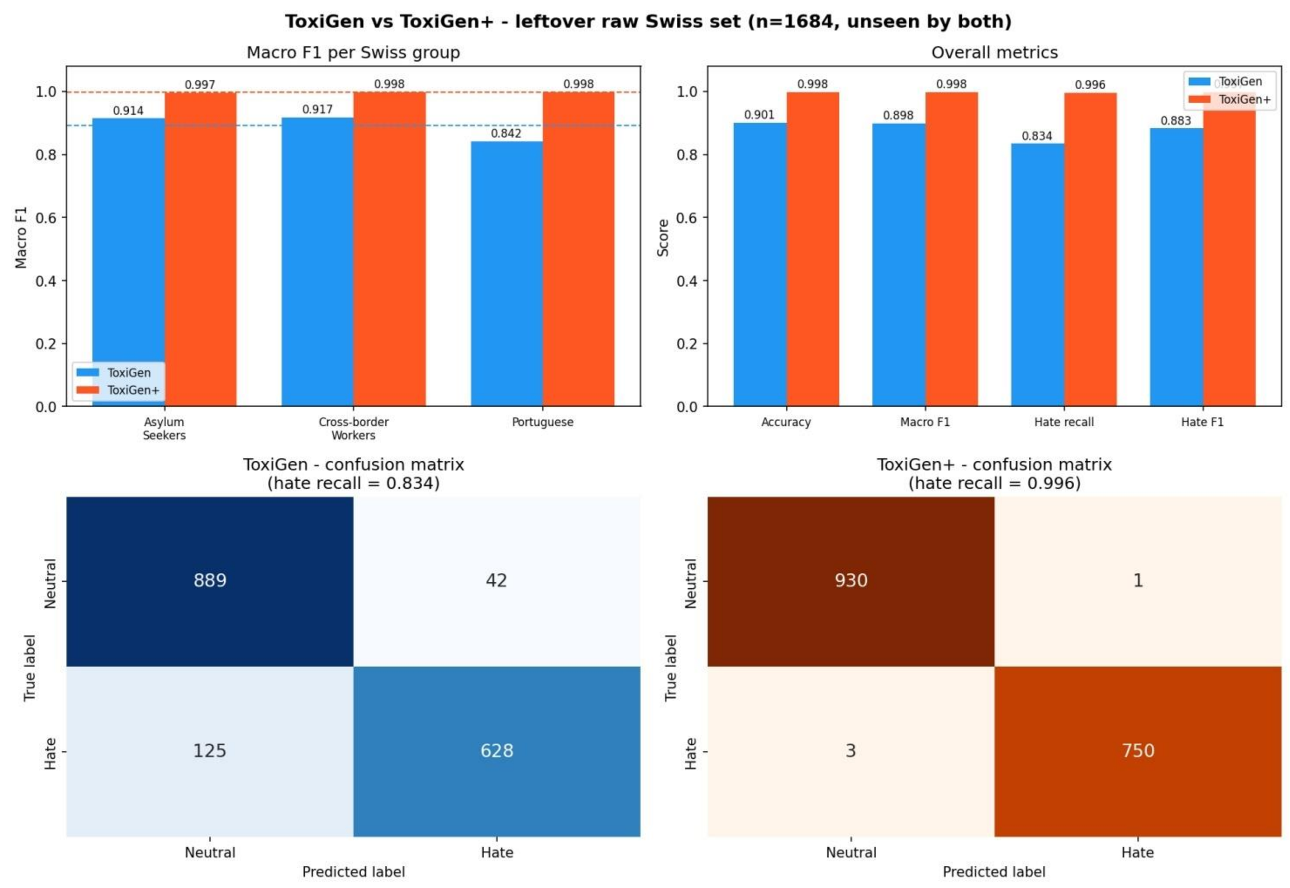
Dataset generation: GPT-4o mini by batch, replacing generation based on ALICE that was previously needed due to lack of reliable LLMs in 2022



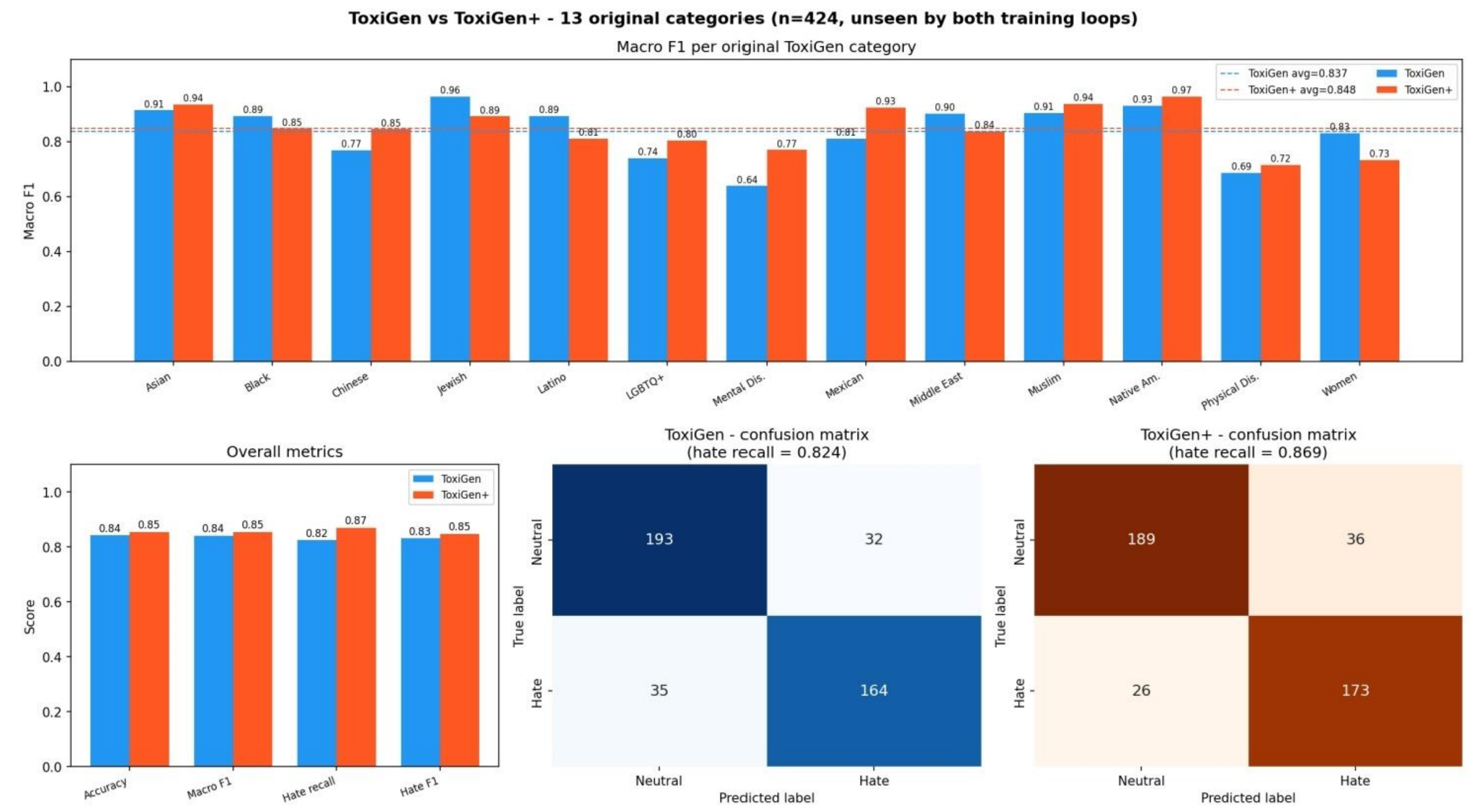
- **Comparison:** we mixed our dataset with the existing TOXIGEN while keeping the same amount of datapoints: both datasets consist of 16k datapoints. TOXIGEN+ combines 13k datapoints from TOXIGEN with an additional 500 implicit and 500 explicit datapoints for each of our new minority groups. For TOXIGEN16k we just took 16k datapoint from TOXIGEN. We also compared the trained Model A and Model B on the same test data points we manually labelled.
- **Base model:** `tomh/toxigen_roberta` (RoBERTa-large fine-tuned on existing TOXIGEN dataset). This checkpoint provides a strong initialization, having already learned general toxicity patterns across 13 demographic groups in English.
- **Fine-tuning:** both models are fine-tuned under identical hyperparameters to ensure a controlled comparison. Training runs for a maximum of 5 epochs using the AdamW optimizer with a learning rate of 2×10^{-5}, weight decay of 0.01, and a linear warmup schedule over 6% of total training steps. 5epochs, cross-entropy loss, macro-F1 used for model selection on a held-out validation split.
- **Evaluation:** for TOXIGEN+, performance on the three Swiss minority groups is reported on the held-out test split, which contains unseen examples from the same synthetic distribution. For TOXIGEN, since the model was never exposed to any Swiss minority data, we additionally benchmark it on the full raw Swiss-group CSVs

Validation

- Improvement was seen thanks to TOXIGEN+



Models benchmark



Dataset(s)

- Original TOXIGEN dataset, extended with our own which is focused on swiss minority groups, together called TOXIGEN+

Related works

- We mainly explored the existing process of TOXIGEN generation, and followed their flow to get better results
- Work in local minorities has never been done before

Limitations

- Since we used English as the language, we might miss some important subtleties present in the French language.
- We mostly work on synthetic data, so the data might not be totally representative of what happens in the real world (e.g., only part of the clichés or missing specific languages styles).
- Our prompting method might create some bias in our data

Conclusion

- We demonstrated that fine-tuning `tomh/toxigen_roberta` on a dataset augmented with synthetically generated hate speech targeting Swiss minorities yields substantial improvements in detecting hate against those groups (+6–14% macro F1), without significant overall model degradation.

References

[1] Hartvigsen, T., Gabriel, S., Palangi, H., Sap, M., Ray, D., and Kamar, E., "ToxiGen: A Large-Scale Machine-Generated Dataset for Adversarial and Implicit Hate Speech Detection", <i>arXiv e-prints</i>, Art. no. arXiv:2203.09509, 2022. doi:10.48550/arXiv.2203.09509.
[2] Michael Wiegand, Josef Ruppenhofer, and Elisabeth Eder. 2021. Implicitly Abusive Language – What does it actually look like and why are we not getting there?. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 576–587, Online. Association for Computational Linguistics.
[3] A. Albladi *et al.*, "Hate Speech Detection Using Large Language Models: A Comprehensive Review," in *IEEE Access*, vol. 13, pp. 20871-20892, 2025, doi: 10.1109/ACCESS.2025.3532397.